

ARTIFICIAL INTELLIGENCE IN ENGINEERING ASSIGNMENT 1 & 2

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Class: CE 3A

Water Jug Problems – State Space Representation Solutions

Question 1: 3-Gallon and 4-Gallon Jugs – Measure Exactly 2 Gallons in 4-Gallon Jug

State Representation: (x, y)

x = amount of water in 3-gallon jug

y = amount of water in 4-gallon jug

Initial State: $(0, 0)$

Goal State: $(x, 2)$ where $y = 2$

State Space Transitions:

$(0,0) \rightarrow \text{Fill 4-gallon jug} \rightarrow (0,4)$

$(0,4) \rightarrow \text{Pour 4} \rightarrow 3 \rightarrow (3,1)$

$(3,1) \rightarrow \text{Empty 3-gallon} \rightarrow (0,1)$

$(0,1) \rightarrow \text{Pour 4} \rightarrow 3 \rightarrow (1,0)$

$(1,0) \rightarrow \text{Fill 4-gallon} \rightarrow (1,4)$

$(1,4) \rightarrow \text{Pour 4} \rightarrow 3 \rightarrow (3,2)$

Goal Achieved: 4-gallon jug contains exactly 2 gallons.

Tabular Representation:

Step	State (3L gallon, 4L gallon)	Action
1	(0, 0)	Start – both jugs empty
2	(0, 4)	Fill the 4-gallon jug from the pump
3	(3, 1)	Pour from 4-gal into 3-gal until 3-gal is full
4	(0, 1)	Empty the 3-gallon jug
5	(1, 0)	Pour the 1 gallon from 4-gal into 3-gal
6	(1, 4)	Fill the 4-gallon jug from the pump again
7	(3, 2)	Pour from 4-gal into 3-gal until 3-gal is full – 2 gallons remain in 4-gal (GOAL)

Question 2: 8L Jug Shared Using 5L and 3L Jugs – Measure Exactly 4 Litres Each

State Representation: (x, y, z)

x = amount in 8L jug

y = amount in 5L jug

z = amount in 3L jug

Initial State: (8, 0, 0)

Goal State: (4, 4, 0)

State Space Transitions:

(8,0,0) → Pour 8L→5L → (3,5,0)

(3,5,0) → Pour 5L→3L → (3,2,3)

(3,2,3) → Pour 3L→8L → (6,2,0)

(6,2,0) → Pour 8L→5L → (1,5,2)

(1,5,2) → Pour 5L→3L → (1,4,3)

(1,4,3) → Pour 3L→8L → (4,4,0)

The goal is reached at step 8. Both the 8-litre jug and the 5-litre jug now contain exactly 4 litres, so the water has been shared equally between the two friends.

Tabular Representation:

Step	State (8-L, 5-L, 3-L)	Action
1	(8, 0, 0)	Start – 8-litre jug is full, others empty
2	(3, 5, 0)	Pour from 8-L into 5-L until 5-L is full

3	(3, 2, 3)	Pour from 5-L into 3-L until 3-L is full
4	(6, 2, 0)	Pour 3-L back into 8-L
5	(6, 0, 2)	Pour remaining 2 litres from 5-L into 3-L
6	(1, 5, 2)	Pour from 8-L into 5-L until 5-L is full
7	(1, 4, 3)	Pour from 5-L into 3-L until 3-L is full – 4 litres remain in 5-L
8	(4, 4, 0)	Pour 3-L back into 8-L – both 8-L and 5-L now have 4 litres (GOAL)